

Diane, 52 · 6 months of nocturnal numbness and tingling in right thumb, index, middle fingers · shakes hand to relieve · **positive Tinel and Phalen signs** · thenar atrophy · EMG: median nerve slowing at the wrist · **carpal tunnel syndrome**

BIO 004 · WEEK 8 · CASE DEEP DIVE

The Peripheral Nervous System

A bookkeeper who wakes at night with numb fingers, a positive Tinel sign, and a median nerve squeezed by its own ligament.

PATIENT CASE

Diane is 52, a bookkeeper. For six months she has been waking up **at 3 AM with numb, tingling fingers — specifically the thumb, index, and middle fingers** of her right hand. She shakes her hand to relieve the symptoms and falls back asleep. Lately her grip is weaker; she dropped a coffee mug last week.

On exam, **tapping over her right wrist crease reproduces tingling into the affected fingers (positive Tinel sign)**; flexing both wrists for 60 seconds reproduces symptoms (**Phalen sign**). There is mild **atrophy of the thenar eminence**. Nerve conduction studies confirm **slowing of the median nerve across the carpal tunnel**. The diagnosis is **carpal tunnel syndrome**.

Diane's case is the peripheral nervous system showing the price of squeezing a nerve into a tight tunnel. To understand it you need to know how the PNS is organized, what a peripheral nerve looks like in cross-section, and the dermatome/myotome maps that link the periphery back to the cord.

GUIDING QUESTION

How is the PNS organized, what protects a peripheral nerve, and why does compression of the median nerve at the wrist produce numbness in some fingers but not others?

1 DIVISIONS

Somatic vs autonomic; afferent vs efferent

● Somatic nervous system

Voluntary control of skeletal muscle; sensation from skin, joints, muscles.

● Autonomic nervous system

Involuntary control of viscera. Sympathetic + parasympathetic + enteric. Covered separately on the ANS page.

Within each, two functional sides: **afferent** (carries information FROM periphery TO CNS) and **efferent** (carries commands FROM CNS TO periphery).

2 SPINAL NERVES

Where the PNS originates

31 pairs of **spinal nerves** exit through intervertebral foramina. Each is a short trunk formed by the union of a dorsal root (sensory) and a ventral root (motor). Immediately after exiting it splits into:

● Dorsal ramus

Innervates muscles and skin of the back at that segmental level.

● Ventral ramus

Innervates muscles and skin of the front, sides, and extremities. In the limb regions, ventral rami combine into *plexuses* (cervical, brachial, lumbar, sacral) that reorganize into named peripheral nerves.

Connective tissue wraps at three levels

A peripheral nerve is a bundle of nerve fibers, just as a muscle is a bundle of muscle fibers.

Three levels of connective tissue wrap:



Endoneurium

Wraps individual axons.



Perineurium

Wraps fascicles (bundles of axons).



Epineurium

Wraps the whole nerve.

Axons can be **myelinated** (each Schwann cell wraps one segment of one axon) or **unmyelinated** (one Schwann cell can support several axons in its cytoplasm, but with no wrapping). Speed of conduction depends on diameter and myelination.

Segmental innervation of skin and muscle

A **dermatome** is the strip of skin innervated by a single spinal nerve. A **myotome** is the group of muscles innervated by a single spinal nerve. Memorizing high-yield levels gives you a bedside map: *C4 shoulder tip; C6 thumb; C7 middle finger; C8 little finger; T4 nipple; T10 umbilicus; L4 medial calf; L5 dorsum of foot, big toe; S1 lateral foot, little toe; S2-S4 perineum.*

Reflexes (single segments): *C5-C6 biceps; C7-C8 triceps; L2-L4 patellar; S1-S2 Achilles.*

Common syndromes by nerve

NERVE	COMMON ENTRAPMENT	FINDINGS
Median	Carpal tunnel	Numbness thumb/index/middle, thenar weakness. <i>Diane.</i>
Ulnar	Cubital tunnel (elbow)	Numbness little/half ring finger, intrinsic hand weakness.
Radial	Spiral groove (humerus)	Wrist drop, can't extend fingers. "Saturday night palsy."
Common peroneal	Fibular head	Foot drop, sensory loss dorsum of foot.

Why the median nerve gets squeezed

The **carpal tunnel** is a narrow passage on the palmar side of the wrist. **Floor:** the carpal bones, arched into a concavity. **Roof:** the *flexor retinaculum* (transverse carpal ligament), a tough fibrous band stretched across the bones. **Contents:** the median nerve plus 9 flexor tendons (4 flexor digitorum profundus, 4 flexor digitorum superficialis, 1 flexor pollicis longus).

10 structures in a tight space with no room to expand. Anything that swells the tendons (repetitive use, rheumatoid arthritis, fluid retention from pregnancy or hypothyroidism) raises pressure inside the tunnel and compresses the median nerve.

Why the symptoms map the way they do. The median nerve provides sensation to the palmar side of the thumb, index, middle, and lateral half of the ring finger — plus motor to the thenar muscles (thumb opposition). Carpal tunnel produces numbness in that specific distribution and weakness of thumb opposition. Crucially, the *palmar cutaneous branch of the median nerve* branches off BEFORE the tunnel, so sensation over the thenar palm is SPARED in carpal tunnel — useful to distinguish from more proximal median lesions.



DRAW ZONE 1 -- Sketch a peripheral nerve in cross-section showing endoneurium, perineurium, epineurium, axons, and myelin.

Use a different color for each wrapping layer.



DRAW ZONE 2 -- *Diagram the carpal tunnel in cross-section at the wrist. Show carpals, flexor retinaculum, 9 tendons, median nerve. Mark Diane's sensory distribution on the hand.*

Median nerve sits between the tendons and the flexor retinaculum — it gets pinched between them when pressure rises.

A tunnel that's too small for what it holds

Carpal tunnel syndrome is the most common nerve entrapment in the body. Diane's symptoms map cleanly to median nerve territory.

"Numbness in thumb, index, middle fingers"

Exactly the median nerve sensory distribution in the hand. Lateral half of ring finger may also be affected.

"Worse at night"

Wrist tends to flex during sleep, narrowing the tunnel and raising pressure on the nerve. Fluid shifts in the recumbent position may also worsen symptoms.

"Shaking the hand helps"

Movement temporarily reduces edema in the tunnel and improves nerve perfusion. The 'flick sign.'

"Positive Tinel sign"

Tapping the inflamed nerve produces tingling in its distribution. Mechanically irritates already-irritable nerve fibers.

"Thenar atrophy"

Chronic median nerve compression denervates the thenar muscles (abductor pollicis brevis, opponens pollicis, superficial flexor pollicis brevis). Weakness of thumb opposition follows. Atrophy = late, often less reversible.

"Treatment"

Wrist splinting at night (keeps wrist neutral), NSAIDs, corticosteroid injection. For severe or progressive symptoms or thenar atrophy: surgical release of the flexor retinaculum (carpal tunnel release).

Spinal nerve, plexus, peripheral nerve, tunnel. Diane's syndrome is the PNS showing how anatomy and physical pressure dictate function.

